

Topic 3: Data

Computers are able to store and manipulate large quantities of data. They use binary to represent different types of data.

Students are expected to learn how different types of data are represented in a computer and should be given the opportunity to gain practical experience of using SQL to handle data stored in a structured database.

Subject content	Students should:
3.1 Binary	3.1.1 Understand that computers use binary to represent data and instructions
	3.1.2 Understand the terms bit, nibble, byte, kilobyte (KB), megabyte (MB), gigabyte (GB), terabyte (TB)
	3.1.3 Be able to convert between binary and denary whole numbers (0-255) and vice versa
	3.1.4 Be able to perform simple binary arithmetic [add, multiply] and understand the concept of overflow
	3.1.5 Understand why hexadecimal notation is used and be able to convert between hexadecimal and binary and vice versa
	3.1.6 Understand that file storage is measured in bytes and data transmission is measured in bits per seconds. Be able to calculate the